

Maxwell Jia

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CAREER OBJECTIVE

As a Master of Computing (Data Science) graduate from the Australian National University, Maxwell is passionate about developing web applications and using AI technologies to solve real-world problems.

Built two web applications and completed an AI internship, Maxwell has hands-on experience with **Spring Boot, Django, ReactJS, data processing, and machine learning**. With a strong passion for technology and excellent technical expertise, Maxwell is seeking a Front end Developer role to create impactful solutions.

SKILLS

- Programming Language: Java, Python, SQL, JavaScript
- Web Development: React, Spring Boot, Django, RESTful, Azure, Axios, MySQL, Junit
- Data Science: Machine Learning, Computer Vision, Audio Processing, Image Processing, Pandas
- Business: Data Analysis, Business Analysis
- Languages: English (Advanced), Chinese (Native)

PERSONAL PROJECTS

User Center Web Page

Project Links: [Online Demo](#), [Front end](#), [Back end](#)

Objective:

Developed a user management system to help the company manage users efficiently. Implemented functions such as registration, encryption, and administrators can add, delete, update, and query users.

Techniques:

- **ReactJS**: The front end is built with React, using an **Ant Design Pro** template to initialize the project.
- **Spring Boot**: The back end uses MySQL for the database, operated via **MyBatis-Plus**, with a Request-Service-Database architecture for maintainability and scalability.
- **Java**: Built a scalable Spring Boot framework in Java and implemented code to handle requests from the front end efficiently, also implemented **Junit** tests to enhance the stability and reliability of the code.
- **Azure**: Azure is used for both front-end and back-end deployment and **MySQL** database services.

Data Inference Web Page

Project Links: [Front end](#), [Back end](#), [Introduction Video](#)

Objective:

This project enables users to upload a CSV file, infer the data types of its columns, and interactively modify these data types as needed if some inferences are not accurate. The file is then processed by the back end to apply the desired data type conversions and an adjusted file will be returned for download.

Techniques:

- **ReactJS**: The front end is built with React, using HTML and CSS for the page design. **JavaScript** handles user interactions, while **Axios** is used to communicate with the back end.

- **Django:** The back end is built with Django, which connects to the front end through URLs. The classes in views call the appropriate methods to handle requests from the front end.
- **Python:** Developed a robust data processing program in Python, leveraging the **Pandas** library for efficient data manipulation and analysis. The program can handle over 20000 data in 20 seconds, perform complex transformations, and ensure data integrity throughout the processing workflow.

INTERNSHIP

AI Machine Learning Engineer Intern

Feb 2024 — Nov 2024

ANU TechLauncher, Canberra

Project Links: [Cam Fall](#)

Objective:

Developed an AI fall detection system using **YOLO7** for real-time video analysis and **ResNet50** for audio spectrogram classification, integrating them to optimize camera fall detection and response.

Job Duties:

- **AI System Development:** Built an AI-based fall detection system using YOLO7 for real-time video analysis and ResNet50 for audio classification, optimizing fall detection accuracy and response.
- **Data Collection and Processing:** Gathered and processed diverse datasets, performing data wrangling and noise reduction to enhance model training quality.
- **Algorithm Implementation:** Deployed YOLO7 for real-time video fall detection and ResNet50 for audio spectrogram classification to accurately recognize fall events.
- **System Integration and Camera Control:** Integrated video and audio detection with real-time camera control via HTTP and an alert system for immediate fall detection notifications.

Main Achievements:

- Achieved an 80% accuracy rate in fall detection by fine-tuning YOLO7 and ResNet50 models, effectively minimizing false positives and enhancing reliability.
- Collected and processed over 1,000 diverse images and audio samples, resulting in a 10% improvement in model performance due to high-quality and well-augmented data.

EDUCATION

Master of Computing (Data Science)

The Australian National University

- High Distinction in Structured Programming Course (Java)
- Transferred from Master of Business Information System

Bachelor of E-commerce

Henan University

- GPA 3.6/4
- National Inspirational Scholarship Winner, 2019
- Outstanding graduates of Henan University

REFERENCES

Other reference available upon request